

Identification and evaluation of the effective criteria for detection of congestion in a smart city

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Abstract

The delay in transportation of necessary items is due to traffic congestion throughout the world. This is a serious phenomenon which results in waste of time and fuel. The detection of road conditions and dissemination of traffic information efficiently and effectively is a big challenge to authorities. Recently, the technologies of vehicular ad hoc networks (VANETs) have been utilized and become an important part of the intelligent transportation system (ITS). For this existing problem, vehicle-to-vehicle (V2V) communication provides a means for cooperation and route management in transport networks. This paper proposed a novel congestion detection system based on the combination of k-means clustering and analytical hierarchy process. In the simulation of urban mobility (SUMO) simulator, a transport network is created and parameters of vehicles facing congestion are taken to extract the key parameter by using the k-means clustering technique and mathematical mean algorithm. This parameter is utilized in analytical hierarchy process to detect the highest priorities parameter and based on that the congestion is detected in particular lane. The result can be a better technique for congestion detection as it requires low installation cost and can be incorporate in vehicles for congestion avoidance which will alternatively improve the traffic flow.

1 | INTRODUCTION

Transportation has a lot of contribution to make this world civilized and developed. With the acceleration of civilization and rapid development of the social economy, the number of vehicles has grown rapidly, which prominently increases traffic congestion and other traffic-related problems. Traffic congestion causes road rage, pollution, and more travel time and fuel consumption [1]. This high congestion spreading issues such as slower speeds, larger travel time, and increased vehicle queuing in urban areas poses a burning requirement to have an effective and adaptive traffic management system. This issue of congestion has affected both the developing and developed

countries in different degrees of aspect depending upon the techniques used and infrastructure available. Traffic congestion mainly occurs when the volume of traffic on a roadway is so heavy that it can force drivers to drive slowly or stop completely. One issue with this congestion is that it is unpredictable and it can happen anywhere depending on the volume of traffic and the road condition that change with time. Different roads have different congestion patterns [2]. Hence an intelligent and practicable system can be designed to overcome congestion.

Many researchers have proposed both infrastructure-based and infrastructure-less solutions to detect the current condition of roads. Some infrastructure-based solutions are vision-based [3], magnetic sensor-based [4] or radio frequency identification

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(RFID) based [5]. These solutions are expensive; require lots of maintenance, have more chance of failure, less accurate.

Infrastructure fewer solutions are global positioning system (GPS), global system for mobile communication (GSM), and accelerometer sensors of smartphones [6]. But priority should be given to approaches that try to predict congestion before the incident of negative effects.

At a junction, when vehicles are facing congestion, nearer vehicles have their parameters very close to each other. So, by using an appropriate clustering algorithm a cluster can be formed with parameters more similar to each other.

In this paper, an ITS is proposed which incorporates the V2V communication in the VANET environment. As the vehicles are dynamic, the popular, fast, robust, and easier clustering technique, that is, k-means clustering using mathematical mean is used to extract the key features from a group of vehicles present in a congested junction area. This k-means clustering technique is easy to implement, adapt to new data set, can handle large, different shapes and sizes data set. The extracted key feature is used further in the analytical hierarchy process to detect the parameter of the message with the highest priority to identify congestion in a particular lane.

The above clustering technique is tested to control the traffic problem in a big city like Bhubaneswar, an administrative, information technology, education, and tourism-driven city. In this paper, the real-world vehicle data are extracted and simulated in Matlab using this congestion technique. The rest of the paper is organized as follows. Section 2 describes some recent related works. Section 3 explains the experimental setup requirements for the proposed model. The proposed model to extract the key features of the vehicle parameters having the highest priority to estimate congestion is described in Section 4. The experimental result is analysed in Section 5. Section 6 explains the statistical analysis of the test data. The scalable architecture is discussed in Section 7. Finally, Section 8 concludes the paper.

2 | RELATED WORK

The road traffic congestion detection technique is classified into two types: traffic congestion detection and control unit and VANET-based congestion detection [7–11]. In traffic congestion detection and control techniques, a variety of sensors are inbuilt in vehicles to gather traffic-related information, and the control part is used to judge the road congestion by examining the data gathered from sensors [12–15]. In the VANET-based congestion detection mechanism, moving vehicles collect the information of the nearby vehicles and exchange the information among the vehicles. The numbers of independent messages are more in existing congestion detection schemes such as congestion recognition based on gain amplifier theory [16], based on wavelet analysis [17], and based on pattern recognition [18]. In [19], authors have explained a structure-free message aggregation scheme where each node vehicle works as a message aggregator. But, the limitation in this paper is that the node vehicles work within a fixed range of the event. Recently, authors have proposed cooperative traffic congestion

detection (CoTEC) technique that uses a fuzzy logic-based message aggregation technique to detect the congestion [20]. In this work, when traffic congestion is detected, warning messages are generated and are disseminated to upstream vehicles to divert the traffics so that vehicles may choose a different alternate path to avoid the congestion. Authors in [21] proposed a machine learning approach to detect congestion based on parameters like hard delay constraints; speed available through GPS vehicle trajectory. The vehicle speed is predicted using Gaussian process in machine learning. Nowadays deep learning techniques have emerged as well-known non-parametric approaches for short-term forecasting, with some models being proposed and tested in various set-ups. Two important literature reviews on deep learning models can be found in [22], and [23]. These show how different deep learning models can be suitable for traffic flow prediction, and why the spatial and temporal correlation in traffic congestion propagation makes the application of such models difficult. In [24] authors measure the disruption due to the severity of traffic network during the morning and evening peak hours using the TomTom data. At different hours of a day, by measuring accessibility authors performed a vulnerability analysis and show the differences of accessibility losses for different links at different times. Authors in [25] suggested a dynamic technique to evaluate the travel times by using the Google Maps API but did not consider population to evaluate accessibility. To analyse dynamic accessibility, authors in [26] calculated travel times between the 584 transport zones defined by the Madrid transport authority using the Google Maps API. Although Google Maps offer a rather easy way to calculate travel times in a city and the option to do so at different times of the day or traffic conditions, they would be a very expensive solution for a large-scale application with thousands of origin-destination couples while they do not provide much detail for the data themselves. In [27], authors detect traffic congestion in VANET using microscopic congestion detection protocol (MCDP) technique. This also helps in estimating vehicles density and assists driver assistance regarding congestion. Authors in [28] proposed a hybrid model using boosted long short-term memory ensemble (BLSTME) and convolutional neural network (CNN) to address the dynamic behaviour of vehicle and to predict congestion in traffic effectively on road. CNN extracts features from traffic images and BLSTME trains and strengthens the weak classifiers for prediction of congestion. This model is having better performance and computational efficiency in predicting the congestion.

In all the existing techniques, traffic congestion detection and information dissemination are integrated. However, in all these above-mentioned papers each of the vehicles using these techniques evaluates traffic congestion independently, which may lead to unreliable or conflicting results due to lack of coordination among the vehicles.

The main contribution of the proposed work is mentioned below.

- The proposed work uses SUMO simulator to extract the parameters of vehicles.

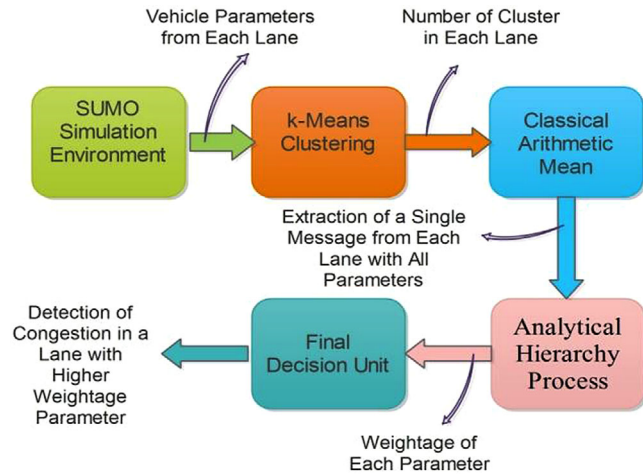


FIGURE 1 Block diagram of the proposed model

- The real-world vehicle data such as speed, fuel consumption, and carbon dioxide (CO₂) emission are extracted by using this simulator.
- It uses the k-means clustering technique to extract the key features from a group of vehicles present in a lane.
- Analytical hierarchy process is used to act on these key features of each lane to priorities the lane congestion based on the priority parameter.
- This proposed work is tested to detect congestion in Bhubaneswar city which is an administrative, education, information technology, and tourism-driven place of Odisha, India.

3 | PROPOSED METHOD

Vehicular ad-hoc network (VANET) is an important component of the intelligent transportation system (ITS). It is used to improve safety and information transmission on roads. The vehicles equipped with wireless communication technologies can communicate with nearby vehicles through V2V communication mode [29, 30] and with fixed infrastructure through V2I and I2V communication [31]. These types of wireless communications are used by vehicles to share information including safety as well as non-safety-related information. In VANET, vehicles are nodes in the network which are highly mobile. During communication with nearby vehicles, lots of information are transmitted which may consume the network bandwidth. To limit the bandwidth requirement, clustering is a better solution. As k-means clustering is a fast, robust clustering technique, it is used here to extract the parameters from a group of messages transmitted between vehicles. The arithmetical mean of these key parameters is calculated for each lane at a junction. Then based on the priority of the parameter with the help of the analytical hierarchy process (AHP), the intensity of traffic congestion is detected in a lane. The block diagram of the proposed model is shown in Figure 1.

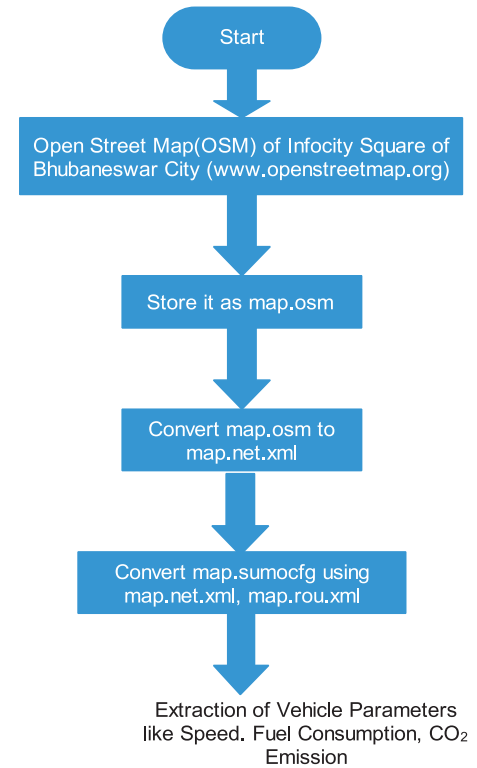


FIGURE 2 Flow diagram for extracting the real-time parameters from vehicles using SUMO simulator

3.1 | Parameter extraction using SUMO simulator

The parameters of vehicles are extracted from a traffic scenario created in the SUMO simulator. The flow chart for extraction of vehicle parameters using the SUMO simulator is shown in Figure 2. The real map of Infocity square, the most crowded junction of Bhubaneswar city near Infocity-1, Patia is taken from the open street map, and from the traffic, the scenario created the parameters of vehicles is extracted by a process in SUMO simulator [32]. It is an open-source, highly portable, microscopic, and continuous road traffic simulation package designed to handle large road networks. The route choice and re-routing facility of SUMO simulator help us to select this in place of Matlab/NS2/TETCOS NetSim.

Figure 3 shows the imported network of Infocity Square taken from <http://openstreetmap.org>. This downloaded map saved in the .osm file is imported to SUMO to create a traffic environment, which is saved in a .cfg file with the help of Netconvert, Polyconvert, and randomTrips.py tools.

The raw outputs such as lane id, carbon monoxide (CO), CO₂, nitric oxide (NO_x), particulate matter (PM_x), noise, fuel consumption, maximum speed, minimum speed etc. can be extracted from the simulator for simulation.

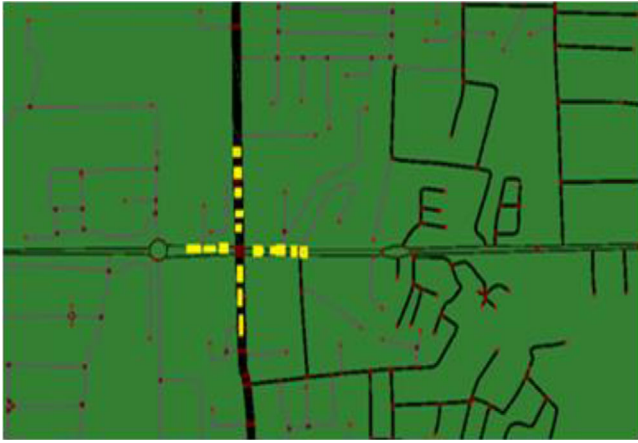


FIGURE 3 Congestion created at Infocity Square, Bhubaneswar [33]

3.2 | K-means clustering technique for congestion detection

When the vehicles are facing congestion some of the parameters are changing from their normal condition. By creating real-time congestion in the SUMO simulator the parameters are extracted and the important parameters which are affected mostly like speed, fuel consumption, CO₂, NO_x are collected. In a congested area as several vehicles are present, to limit the bandwidth required for transmission of individual vehicle parameters, the clustering technique is used to extract a single message in place of many.

k-means clustering is a fast, robust, portioning technique where all messages of the data set DSUMO are organized into k groups/clusters, where $k \leq m$. The k-means algorithm is mentioned below [34, 35].

Input:

- k : the number of clusters
- D : the data set containing m objects

Output:

- A set of k clusters

Method:

1. k objects from the dataset D are selected arbitrarily as the initial cluster centres;
2. The distance between each data point and cluster centre is calculated;
3. The data point is assigned to the cluster centre whose distance from the cluster centre is the minimum of all the cluster centres;
4. The cluster centre is recalculated;
5. The distance between each data point and the new obtained cluster centre is recalculated;
6. If no data point is reassigned then stop otherwise repeat from Step (3).

The data set DSUMO which indicates the messages with features of speed (km/hr), CO₂ (mg/s) emission, and fuel consumption (mL/s) are collected from the SUMO Simulator and is listed in Table 1.

Based on the closeness of parameters the clusters are formed using k-means clustering for a particular lane. The key parameters for that lane are the mean value of the centres of the clusters.

Let $X_1, X_2, X_3, \dots, X_n$ is the centre of n number of clusters, then the mean value of speed, fuel consumption, and CO₂ emission are estimated for a set of data by using the Equations (1)–(3).

$$S_m = \frac{\sum_{i=1}^n S_i}{n} \quad (1)$$

$$F_m = \frac{\sum_{i=1}^n F_i}{n} \quad (2)$$

$$C_m = \frac{\sum_{i=1}^n C_i}{n} \quad (3)$$

where S_i, F_i , and C_i are the speed, fuel consumption, and CO₂ emission of i th centre X_i .

S_m, F_m , and C_m are the mean values of speed, CO₂ emission, and fuel consumption of all centres.

3.3 | Analytical hierarchy process (AHP)

Analytic hierarchy process (AHP) was introduced by Saaty [36, 37] as a powerful tool for decision making. This is an Eigenvalue approach to pair-wise comparisons. AHP allows the decision-maker to make comparisons of choices based on various criteria through his or her judgment. These criteria may or may not have to be quantifiable. This method is based on three principles:

- Problem identifications and model structure preparation
- Comparative judgment of the criteria and the alternatives
- Evaluation of relative weight

Here the decision-maker will do a pair-wise comparison in a matrix-based on the following scale introduced by Saaty [36, 37] in Table 2.

The pair-wise judgment starts with a comparison of criteria according to their levels of influence and is based on the specified criteria in the higher level. The pair-wise comparison matrix (n -by- n) will be derived by using Equation (4).

$$\begin{bmatrix} 1 & x_{11} & x_{12} & \dots & x_{1n} \\ 1/x_{11} & 1 & \dots & \dots & \vdots \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ 1/x_{n1} & \dots & \dots & \dots & 1 \end{bmatrix} \quad (4)$$

TABLE 1 Data set DSUMO

Number of vehicles	Speed (km/h)	Lane 1	
		Fuel consumption (mL/s)	CO ₂ emission (mg/s)
1	0	1.13	2624.72
2	1.66	1.53	3127.82
3	4.36	2.17	4885.96
4	27.27	3.73	8884.93
5	9.30	3.52	7360.24
6	6.10	2.18	5061.23
7	3.41	1.49	3396.26
8	27.46	2.55	5478.48
9	8.59	3.67	7377.97
10	4.78	2.14	4744.79
11	2.54	1.27	2941.96
12	21.38	2.39	5345.66
13	11.72	3.69	8454.58
14	26.62	2.12	4925.89
15	23.21	3.57	8555.74
16	6.75	2.48	5544.28
17	7.77	2.19	5172.06
18	6.55	2.72	6232.26
19	27.51	4.39	10,215.84
20	8.85	1.88	4456.62
21	7.24	2.41	5845.94
22	27.31	4.59	10,115.84
Number of vehicles	Speed (km/h)	Lane 2	
		Fuel consumption (mL/s)	CO ₂ emission (mg/s)
1	0.86	1.25	2622.32
2	1.70	1.77	3180.88
3	2.52	1.34	2753.50
4	3.32	1.79	3657.93
5	9.35	3.65	8482.62
6	11.43	3.03	6983.76
7	12.56	3.66	8771.36
8	16.68	3.87	9041.59
9	17.73	4.76	11,371.44
10	21.34	7.64	18,016.88
11	22.98	7.62	17,690.26
12	8.46	2.73	6577.98
13	26.13	7.85	18,135.66
14	18.32	4.89	11,247.94
15	12.81	4.44	10,653.88
16	5.74	2.42	5684.19
17	6.56	1.79	4417.81
18	28.64	3.77	8649.24
19	19.78	7.12	16,340.06

(Continues)

TABLE 1 (Continued)

Number of vehicles	Speed (km/h)	Lane 2	
		Fuel consumption (mL/s)	CO ₂ emission (mg/s)
1	3.11	1.54	3555.93
2	26.65	8.66	20,262.73
3	6.81	2.14	4795.25
4	9.43	3.65	8782.62
5	12.57	3.88	8978.37
6	4.28	2.19	5016.74
7	14.17	5.93	11,790.13
8	6.28	2.75	5336.79
9	27.69	4.45	10,566.67
10	8.76	2.85	6588.98
11	17.78	6.22	14,213.07
12	24.52	5.37	12,486.57
13	10.97	2.58	5993.99
14	21.93	5.74	13,892.30
15	12.75	3.32	7738.81
16	15.16	5.34	11,797.48
Number of vehicles	Speed (km/h)	Lane 4	
		Fuel consumption (mL/s)	CO ₂ emission (mg/s)
1	6.72	2.23	4795.25
2	9.32	3.62	8482.62
3	11.55	3.87	8981.37
4	19.41	6.45	14,913.19
5	27.79	4.37	10,486.67
6	24.15	7.74	17,656.55
7	8.56	2.76	6588.76
8	17.95	6.52	14,113.07
9	20.20	5.76	13,918.78
10	10.67	2.68	5963.98
11	21.83	5.53	13,392.30
12	23.19	5.13	11,973.11
13	15.04	5.12	11,677.48
14	0.1	1.03	2628.65
15	5.3	1.98	4098.3

Here α is the intensity of importance of one criterion concerning another.

This $\alpha_{ii} = 1$ for $i = 1, 2, \dots, n$;

$\alpha_{ij} = 1/\alpha_{ji}$ for $i = j = 1, 2, \dots, n$; with $i \neq j$

TABLE 2 Saaty's intensity of importance scale

Intensity of importance	Definition explanation	Definition explanation
1	Equal importance	Two activities contribute equally to the objective
3	Moderate	Importance experience and judgment slightly favour one activity over another
5	Strong importance	Experience and judgment strongly favour one activity over another
7	Very strong importance	Activity is strongly favoured and its dominance demonstrated in practice
9	Extreme importance	The evidence favouring one activity over another is of the highest possible order of affirmation
2,4,6,8	Intermediate values between two adjacent judgments	When compromise is required
Reciprocals: 1/3, 1/2, 1/7, 1/8, etc.	Reciprocal values	Inverse comparison

To calculate the weightage of each of the criteria during the selection of a choice normalized pair-wise matrix is required to be calculated.

The normalized pair-wise matrix (n -by- n) will find out by using Equation (5).

$$\begin{bmatrix} \frac{1}{\sum_{i=1}^n 1/x_{i1}} & \cdots & \frac{x_{1n}}{\sum_{i=1}^n x_{1n}} \\ \vdots & \ddots & \vdots \\ \frac{1/x_{n1}}{\sum_{i=1}^n 1/x_{i1}} & \cdots & \frac{x_{nn}}{\sum_{i=1}^n x_{nn}} \end{bmatrix} \quad (5)$$

As humans are unable to make perfect judgments, some degree of inconsistency in these comparisons may occur. There is a need to perform consistency verification to ensure that judgments stay within an acceptable range of inconsistency [38]. This is done by adding the columns in the judgment matrix and multiplies the resulting vector by the vector of priorities obtained earlier. The consistency index (CI), that Saaty proposed, needs to be first calculated by using Equation (6) [37]:

$$CI = \frac{\lambda_{\max} - n}{n - 1} \quad (6)$$

where n = number of compared elements (criteria), that is, the size of the matrix.

$\lambda_{\max}$ = the real dominant maximum Eigenvalue.

Using the CI, the consistency ratio (CR) can be then derived by using Equation (7) [37]:

$$CR = \frac{CI \text{ of } A}{RI \text{ for size } n} \quad (7)$$

TABLE 3 Random index value

Size of matrix (n)	Random index (RI)
1	0.00
2	0.00
3	0.58
4	0.90
5	1.12
6	1.24
7	1.32
8	1.41
9	1.45
10	1.49

where RI is the random index which can be acquired from Table 3 [38].

The rule of thumb is to have the value of CR be under 0.1, which equates to a 10% inconsistency [38]. This gives us the idea that the matrix is constant. That means the criteria having the highest weightage from this matrix can be chosen for the selection process.

4 | EXPERIMENTAL SETUP

For performing the analysis of the proposed system, an experimental setup was prepared. For creating the traffic environment at Infocity square of Bhubaneswar City, Odisha, India with Latitude and longitude coordinates 20.344744 and 85.803818 respectively, SUMO Simulator 1.9.1 is used for Windows 10 64bit. It is a portable, microscopic road traffic simulation package. It is developed by the DLR Institute of Transportation Systems. For making clusters of the vehicles at Infocity Square, Matlab 2015 is used [39] in Asus Vivo Book Intel 14 Core i3 10th Gen-4GB CPU @ 2.1 GHz with Turbo Boost up to 4.1 GHz with 4GB RAM computer.

5 | RESULT ANALYSIS

The k-means clustering technique is doing clustering of different information gathered from all variety of vehicles in a lane of Infocity Junction. The function k-means works on the data set generated and filled by information coming from the vehicles and finally, some clusters are generated.

A three-dimensional plot formed with three attributes, speed, fuel consumption, and CO₂ emission for all the vehicles is shown in Figures 4–7. Here black bubble marks show the centres of the clusters. The three cluster centres and their mean value for each lane are mentioned in Tables 4–7.

The mean value of the cluster centres is the key message from this group of data. The key parameters of vehicles of each lane are listed in Figure 8 below.

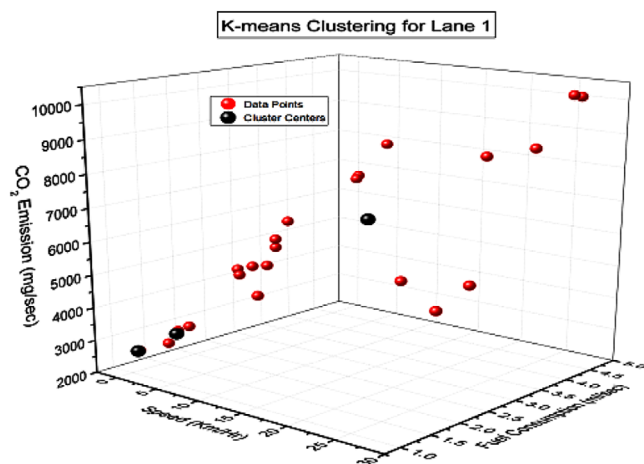


FIGURE 4 Formation of clusters taking data messages in Lane 1

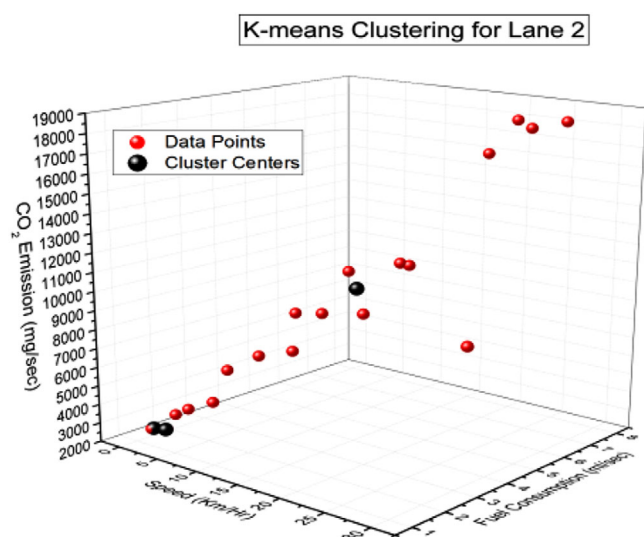


FIGURE 5 Formation of clusters taking data messages in Lane 2

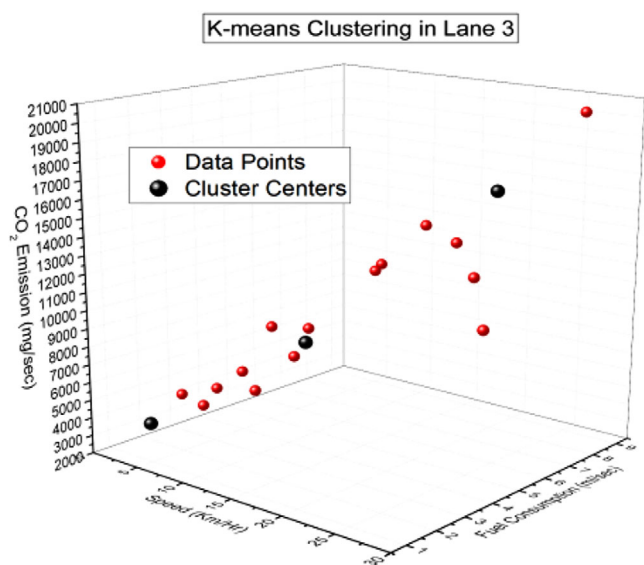


FIGURE 6 Formation of clusters taking data messages in Lane 3

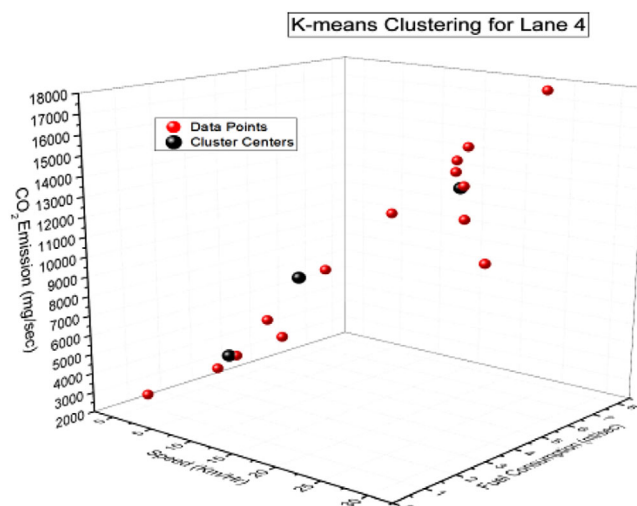


FIGURE 7 Formation of clusters taking data messages in Lane 4

TABLE 4 Centres and mean of the centres' parameters of the clusters in Lane 1

Clusters	Speed (km/h)	Fuel consumption (mL/s)	CO ₂ emission (mg/s)
Cluster 1	0	1.1	2624.7
Cluster 2	2.5	1.4	3155.3
Cluster 3	14.6	2.9	6592.1
Mean value	5.7	1.8	4124.03

TABLE 5 Centres and mean of the centres' parameters of the clusters in Lane 2

Clusters	Speed (km/h)	Fuel consumption (mL/s)	CO ₂ emission (mg/s)
Cluster 1	0.9	1.3	2622.3
Cluster 2	14.3	4.3	9935.5
Cluster 3	2.5	1.3	2753.5
Mean value	5.9	2.3	5103.77

We have identified three criteria which we propose to be important for congestion detection. These criteria are selected depending on the parameters of vehicles extracted from SUMO simulator and are able to fulfil the needs of congestion detection evaluation. The criteria we propose here are: speed, fuel emission and CO₂ emission. Figure 9 summarizes the meaning of each of the criteria.

The weight of each criterion for congestion detection is calculated by using the model AHP. The criterion in accordance with our interest and the sample of the pair-wise comparison matrix is mentioned in Table 8.

Table 8 shows the pair-wise comparison matrix consists of:

- The comparison value of each criterion with itself is 1, which means the intensity of the same interest.

TABLE 6 Centres and mean of the centres' parameters of the clusters in Lane 3

Clusters	Speed (km/h)	Fuel consumption (mL/s)	CO ₂ emission (mg/s)
Cluster 1	3.1	1.5	3555.9
Cluster 2	22.1	7.2	16,123.1
Cluster 3	12.8	3.7	8321.9
Mean value	12.67	4.13	9333.63

TABLE 7 Centres and mean of the centres' parameters of the clusters in Lane 4

Clusters	Speed (km/h)	Fuel consumption (mL/s)	CO ₂ emission (mg/s)
Cluster 1	0	1.1	2624.7
Cluster 2	2.5	1.4	3155.3
Cluster 3	14.6	2.9	6592.1
Mean value	5.7	1.8	4124.033

- Speed of vehicles in comparison to fuel consumption while facing congestion is 9 means speed is more important than fuel consumption by vehicles.
- Speed of vehicles in comparison to CO₂ emission by vehicles facing congestion is 7 means is more important than CO₂ emission.

- Fuel consumption in comparison to speed and CO₂ emission is 1/9 and 1/5 respectively as are less important from them.
- CO₂ emission in comparison to speed and fuel consumption is 1/7 and 5 respectively.

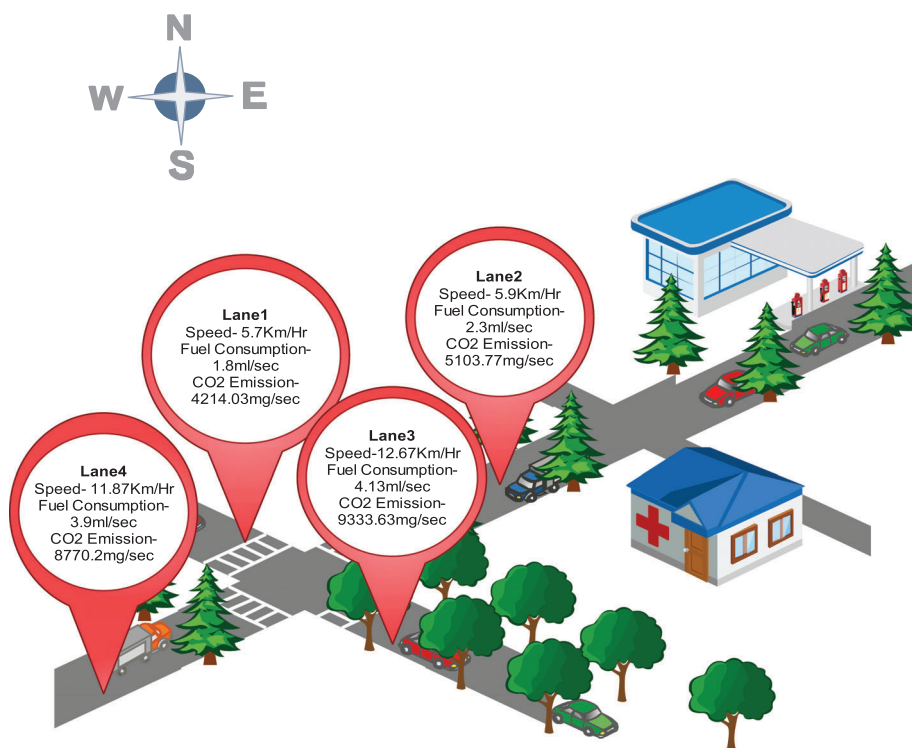
The Eigenvalue for this pair-wise comparison matrix is 3.0194 with CI as 0.0097 and CR as 0.0167. As CR is 0.0167, that means inconsistency is within 10%. The corresponding weight of the three parameters of this matrix consisting of the mean of cluster heads of each lane is calculated and given in Equation (8):

$$\begin{pmatrix} 0.7504 \\ 0.0599 \\ 0.1898 \end{pmatrix} \quad (8)$$

From Equation (8), it is cleared that speed of vehicle is having highest weightage.

The speed, fuel consumption, and CO₂ emission for all four lanes are summarized in Figure 10.

The threshold values of the plots in Figure 10 are the mean values of the cluster heads as discussed in Tables 4–7. According to the figure, as all the parameters speed, fuel consumption and CO₂ emission of Lane 3 are higher than all other lanes, Lane 3 can be considered as a congested lane and necessary steps can be taken to dismiss congestion.

**FIGURE 8** Key parameters of vehicles of each lane

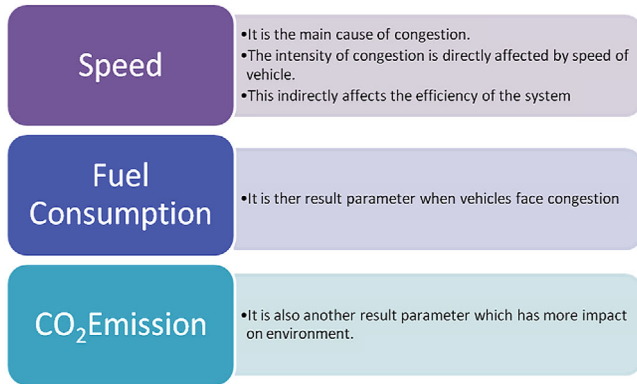


FIGURE 9 Criteria and their explanation

TABLE 8 Pair-wise comparison matrix criteria

Criteria	Speed	Fuel consumption	CO ₂ emission
Speed	1/1	9/1	7/1
Fuel consumption	1/9	1/1	1/5
CO ₂ emission	1/7	5/1	1/1

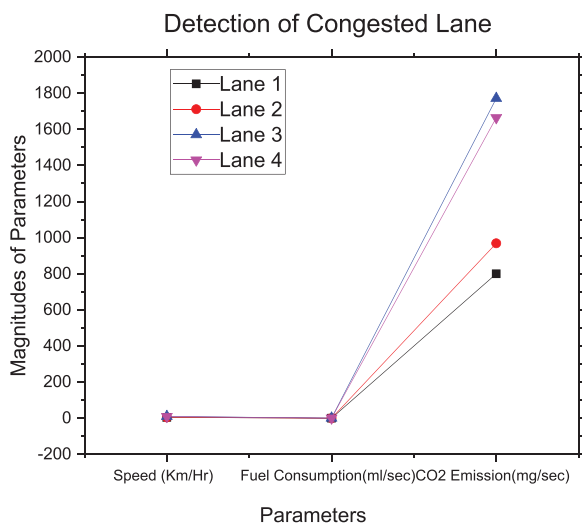


FIGURE 10 Input data to detect congested lane

6 | STATISTICAL ANALYSIS

The variability of the test data for each lane is validated by statistical analysis. From Equation (8) it is cleared that

- The weightage of the parameter “speed of vehicles” is higher than other two parameters
- That’s why speed of vehicles is chosen for statistical analysis

From the Table 9, it is cleared that in Lane 3 the standard deviation calculated is 7.59. As it is less than standard deviation of all other lanes, it is concluded that vehicles are closer to each

TABLE 9 Statistical analysis of each lane data

	Lane 1	Lane 2	Lane 3	Lane 4
No. of vehicles	22	19	16	15
Mean speed	12.29	12.99	13.93	14.79
Standard deviation (SD)	9.67	8.29	7.59	7.77
Standard error of the mean (SEM)	2.062	1.901	1.898	2.005

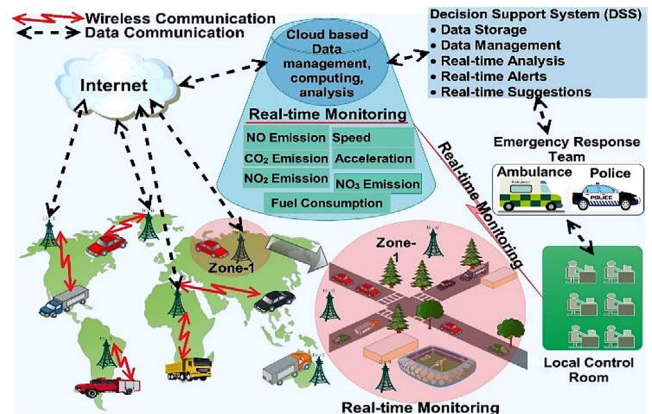


FIGURE 11 Scalable architecture of the proposed technique

other. This statistical analysis justifies that the Lane 3 is congested.

7 | SCALABLE ARCHITECTURE

The scalable architecture of the proposed identification and evaluation of the effective criteria for the detection of traffic congestions techniques is portrayed in Figure 11. The architecture provides a detailed procedure to evaluate the traffic congestions by effectively collecting real-time vehicle parameters in a smart city through 2G/3G/4G/5G communication channel. Further, the vehicles are complex automobile systems and the current automobiles are hybrid systems in conjunction with an advanced embedded system known as electronic control unit (ECU). In the proposed architecture, the real-time sensor data is accessed by the ECU and the auxiliary communication node sends the information to the cloud system for necessary processing. The real-time data collection can be performed in a cloud computing platform to perform data management, computing, and analysis etc. in a large-scale distributed congestion monitoring scenario. Location-specific local control rooms can be established to monitor traffic congestions in a real-time fashion. A decision support system (DSS) will be responsible to process the real-time sensor information and produce adequate emergency alerts for the response teams. In the similar context, this DSS model can also correlate with the traffic congestion values all over the city and produce a shortest route for the emergency vehicles whenever required. Various shortest path models are proposed in several research articles. Some of the shortest path estimation approaches like travelling

salesman problem (TSP), ant-colony optimization models and fuzzy Topsis based models can be used in-conjunction with the DSS model [40–45]. The proposed distributed vehicular ad-hoc network (VANET) will connect the vehicles throughout a city and estimate the traffic congestion.

8 | CONCLUSION

The traffic congestion problems in urban areas of all countries of the whole world are very serious and have become more complicated over the years. Many factors decide the intensity of this problem, like the speed of the vehicle, fuel consumed by vehicles near the congested area, CO₂ emission from vehicles. In this paper, an intelligent congestion detection technique is proposed using k-means clustering and an analytical hierarchy process. The traffic data are collected from SUMO Simulator which imported the real map of Infocity square, Bhubaneswar from open street map. As it is connecting the official, commercial as well as residential areas, the junction needs to be cleared in a timely manner before congestion happens in a lane. At the junction, the vehicles present in a lane are forming clusters depending on the closeness of their parameters, and a single key parameter is extracted with the highest priority. As the consistency ratio is calculated to be 1.67%, the speed of vehicles is taken as the priority parameter for congestion detection. Base on vehicle speed, lane 3 is detected as a more congested area. This information can be further used for controlling the time duration as well as the sequence of operation of traffic lights present at a junction and also can be communicated to drivers moving towards a congested junction during peak hours.

CONFLICT OF INTEREST

The authors wish to confirm that there are no known conflicts of interest associated with this publication and there has been no significant financial support for this work that could have influenced its outcome.

DATA AVAILABILITY STATEMENT

Data available on request from the authors.

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